
IAL Chemistry Unit 5

Rapid Revision

WCH15 — Transition Metals and Organic Nitrogen Chemistry

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* guide (20 past papers, Jun 2019 - Jun 2026)

90 marks · 1 h 45 min · ≈ 1.17 min per mark

Section A: 20 MCQ (**the paper itself says ≤ 20 minutes**) · Section B: ~ 50 · **Section C: ~ 20 marks in ONE context question**

1–3 asterisked (*) questions

TRIAGE — WHERE THE MARKS ARE

Topic	Marks/paper	Frequency
Transition metal complexes: shapes, ligands, colour, reactions	15-25	Every paper
Redox: oxidation numbers, half-equations, titration calculations	12-20	
Arenes: benzene structure, electrophilic substitution mechanisms	10-18	
Electrode potentials, cells, feasibility	8-17	
Organic nitrogen: amines, amides, amino acids, diazonium	8-16	
Spectroscopy and structure determination	5-14	~90%
Organic synthesis routes (Grignard, nitriles)	5-12	~85%
Catalysis (hetero-, homo-, autocatalysis)	3-8	~75%
Polymers	3-10	~60%

TWO PLANNING CONSEQUENCES

- 1. Transition metals plus redox and electrochemistry is 35-50 marks — over half the paper.** If your inorganic is weak you cannot reach an A*, however good your organic is.
- 2. Section C is 20 marks in one question**, always context-based and **synoptic**, drawing on Units 1-4. Never leave yourself short of time for it.

In every single paper: the **colour-of-complexes explanation** · a **redox titration calculation** · an **electrophilic substitution mechanism** · an **E°cell feasibility argument**. **Colour of complexes and aromatic-reactivity comparisons together are more than half of all * questions.**

SHEET 1 — OXIDATION NUMBERS & HALF-EQUATIONS

Rules: uncombined element 0 · simple ion = its charge · Group 1 +1, Group 2 +2, Al +3 · H +1 **except –1 in metal hydrides** · O –2 **except –1 in peroxides, +2 in F₂O** · **F always –1** · sum = 0 (compound) or the charge (ion).

In complexes, work out the charge on the METAL, not on the whole ion. Neutral ligands (H₂O, NH₃, CO, en) contribute 0; anionic ligands (Cl[–], CN[–], OH[–], SCN[–], C₂O₄^{2–}, EDTA^{4–}) contribute their own charge.

[Cr(OH)₆]^{3–} → Cr + 3 · Cr₂O₇^{2–} → +6 · K₂FeO₄ → Fe +6 · [Fe(CN)₆]^{4–} → Fe +2 · VO₂⁺ → V +4 · VO₂⁺ → V +5 · [Rh(CO)₂I₂][–] → Rh +1.

PROVING A REACTION IS REDOX — THE 2-MARK ANSWER, EVERY TIME

"[Element] changes from [number] to [number], so it is oxidised; [other element] changes from ___ to ___, so it is reduced. Both occur, so it is a redox reaction."

Quote the numbers, not just "it gains electrons" — the schemes require them.

Proving NOT redox: 2CrO₄^{2–} + 2H⁺ → Cr₂O₇^{2–} + H₂O — chromium is +6 on both sides, O is –2 and H is +1 throughout, so **no oxidation numbers change**; it is an **acid-base equilibrium**.

Building a half-equation: write the species that change → balance the changing atom → **balance O with H₂O** → **balance H with H⁺** (in alkali, add OH[–] to both sides) → **balance charge with e[–]**. To combine: **reverse the oxidation half-equation**, multiply so the electrons cancel, add and cancel.

Disproportionation: 2Cu⁺ → Cu²⁺ + Cu · **3MnO₄^{2–} + 4H⁺ → 2MnO₄[–] + MnO₂ + 2H₂O** (acid) · 3MnO₄^{2–} + 2H₂O → 2MnO₄[–] + MnO₂ + 4OH[–] (alkali). To build one: take the half-equation where the species is **reduced** and the **reverse** of the one where it is **oxidised**.

Stoichiometry from oxidation numbers: find the change **per atom** → × number of atoms → **total electrons lost = total gained**.

SHEET 2 — ELECTRODE POTENTIALS & CELLS

THE SHE — FOUR CONDITIONS, ALL NEEDED

298 K · H₂ at 100 kPa · [H⁺] = 1.00 mol dm^{–3} · platinum electrode coated with platinum black.

Why platinum black: large surface area and it catalyses the equilibrium between H₂ and H⁺, so the potential is reached quickly and is stable — **not "to protect the electrode"**. **Why platinum:** inert and conducts.

Half-cells: metal/metal ion → the metal itself in 1.00 mol dm^{–3} of its ion · two aqueous ions → **platinum** with 1.00 mol dm^{–3} of **both** · ion + gas → Pt, gas at 100 kPa · ion + H⁺ (MnO₄[–]/Mn²⁺) → **Pt with 1.00 mol dm^{–3} of MnO₄[–], Mn²⁺ AND H⁺**.

The most common lost mark: forgetting that acidified half-cells need 1.00 mol dm^{–3} H⁺ as well.

Salt bridge: saturated KNO₃ (KCl only if no Ag⁺ or Pb²⁺). Function: **completes the circuit by allowing ions to flow, maintaining electrical neutrality. Not a wire**, because a wire lets electrons flow directly and the metal would react / set up its own potentials.

CELL DIAGRAMS — FIVE CONVENTIONS

1. The **most negative** electrode system on the **LEFT** · **2.** single line | = phase boundary · **3.** double line || = salt bridge · **4.** same phase separated by a comma · **5.** working outward from the bridge, the **oxidised species is nearest the bridge**.

Fe(s) | Fe²⁺(aq) || Sn²⁺(aq) | Sn(s) · Pt | I[–], I₂ || S₂O₈^{2–}, SO₄^{2–} | Pt. **Platinum is needed on both sides whenever neither half-cell contains a solid metal.**

FEASIBILITY — AND THE THREE RELATIONSHIPS

E°cell = E°(more positive) – E°(more negative). A reaction is **thermodynamically feasible if E°cell is positive**.

ΔS_{total} = nFE°cell / T · ΔG = –nFE°cell · ln K = ΔS_{total} / R

So **E°cell is directly proportional to ΔS_{total} and to ln K — not to ΔS_{system} and not to K itself**. That exact distractor set recurs in Section A.

THE CRITICAL LIMITATION — a guaranteed point: a feasible reaction may still not occur because **(1) the activation energy may be very high, so the rate is negligible**, or **(2) the conditions are not standard**.

S₂O₈^{2–} + 2I[–] is feasible but slow because **both ions are negatively charged and repel**, giving a high E_a.

Effect of concentration: write the half-equation as an equilibrium → decide which way it shifts on dilution → **shift right (towards reduction) = E more positive; shift left = more negative** → then work out the effect on E°cell. Nernst (given when needed): **E = E° + (RT/nF) ln[Mⁿ⁺]**, F = 96 500.

Running the cell: at the **more positive (right) electrode reduction occurs**, so the oxidised species falls and the reduced rises; a deposited metal **gains mass**; E°cell **falls towards zero**. **Why the reading differs from the Data Booklet: non-standard conditions**, impure solutions, or equilibrium not yet established.

Calomel electrode (Hg₂Cl₂ + 2e[–] ⇌ 2Hg + 2Cl[–], +0.24 V): **KCl crystals keep the solution saturated so [Cl[–]] stays constant. Advantages over the SHE:** safer (no flammable hydrogen), portable, easier to set up, no controlled gas supply.

Fuel cell: acidic — anode **H₂ → 2H⁺ + 2e[–]**, cathode **O₂ + 4H⁺ + 4e[–] → 2H₂O**; alkaline — H₂ + 2OH[–] → 2H₂O + 2e[–], O₂ + 2H₂O + 4e[–] → 4OH[–]. **E°cell = +1.23 V.**

Oxidation at the anode, which is the negative electrode. Advantages: only water at the point of use; high efficiency (**chemical directly to electrical, no heat step**); runs continuously. Disadvantages: hydrogen is **hard and hazardous to store**; usually **made from fossil fuels**; expensive Pt catalysts; little infrastructure.

SHEET 3 — REDOX TITRATIONS (4-6 marks, every paper)

System	Indicator	End point
MnO ₄ ⁻ with Fe ²⁺ , C ₂ O ₄ ²⁻ , H ₂ O ₂	Self-indicating	colourless → first permanent pale pink
Cr ₂ O ₇ ²⁻ with Fe ²⁺	Needs one — diphenylamine sulfonate	green → purple/violet
Iodine-thiosulfate	Starch, added when pale straw/yellow	blue-black → colourless

THE THREE "WHY" ANSWERS

Why dichromate needs an indicator: the **orange → green change is not sharp enough** — the green obscures the end point.

Why starch is added late: at high [I₂] the **starch-iodine complex is so stable that iodine is adsorbed and released only slowly**, giving a late/inaccurate end point.

Which acid: dilute SULFURIC only. **Not HCl** — Cl⁻ is oxidised to Cl₂, giving a **high titre**. **Not HNO₃** — itself an oxidising agent. **Not ethanoic** — too weak, [H⁺] too low.

Insufficient acid: MnO₄⁻ is reduced only to **MnO₂, a brown suspension** — a **3-electron rather than 5-electron** change, so **more MnO₄⁻ is needed and the titre is higher**.

KMnO₄ advantages/disadvantages: self-indicating and sharp; but **not a primary standard** — hygroscopic, impure, **decomposes slowly in solution**, so it must be standardised; the dark colour makes the meniscus hard to read.

The universal layout — six lines, each a mark: **1.** moles of titrant ($n = cV/1000$) · **2.** mole ratio from the equation · **3.** moles in the aliquot · **4.** moles in the full flask ($\times 250.0/25.0$) · **5.** mass or concentration · **6.** percentage, to an **appropriate number of significant figures**.

Ratios: MnO₄⁻ : Fe²⁺ = **1:5** · MnO₄⁻ : C₂O₄²⁻ = **2:5** · Cu²⁺ : S₂O₃²⁻ = **1:1** (via 2Cu²⁺ + 4I⁻ → 2CuI + I₂, then I₂ + 2S₂O₃²⁻) · Cr₂O₇²⁻ : S₂O₃²⁻ = **1:6**.

Back titration: total moles of oxidant — moles remaining = moles reacted → ratio → mass and %.

Water of crystallisation: take 100 g → mass of anhydrous part and of water → moles of each → divide by the smaller → the ratio gives *n*.

ppm/ppb: 1 ppm by mass = 1 g in 10⁶ g = **1 mg dm⁻³** for a solution of density 1.0 g cm⁻³.

pV = nRT in **Pa, m³, K**; molar volume at r.t.p. 24.0 dm³ mol⁻¹. **Gaseous hydrocarbon combustion:** volumes are in the same ratio as moles; **contraction on cooling** excludes the water (it condenses); **CO₂ is absorbed by KOH**, so the further decrease gives V(CO₂).

SHEET 4 — TRANSITION METALS: DEFINITIONS & CONFIGURATIONS

THE TWO DEFINITIONS — AND THE ION/ATOM TRAP

d-block element: an element whose **highest-energy electron enters a d subshell**.

Transition element: a d-block element that forms **at least one stable ION with a partially (incompletely) filled d subshell**.

So **Sc** (only Sc³⁺, 3d⁰) and **Zn** (only Zn²⁺, 3d¹⁰) are d-block but **not** transition elements.

The answer must be in terms of the ION, not the atom. "Zinc has a full 3d subshell" does **not** score; **"the Zn²⁺ ion has a completely full 3d subshell"** does.

Configurations: Sc [Ar]3d¹4s² ... **Cr** [Ar]3d⁵4s¹ ... Mn 3d⁵4s² ... Fe 3d⁶4s² ... **Cu** [Ar]3d¹⁰4s¹ ... Zn 3d¹⁰4s². **Cr and Cu are the exceptions** — a half-full or full d subshell is more stable.

Forming ions: the 4s electrons are removed FIRST. Fe²⁺ [Ar]3d⁶ · Fe³⁺ [Ar]3d⁵ · Cu²⁺ 3d⁹ · Cr³⁺ 3d³.

Why variable oxidation states: the **4s and 3d orbitals are very close in energy**, so successive ionisation energies increase only gradually — **no large jump** — so different numbers of electrons can be removed, and the energy released on bonding or hydration compensates.

Maximum oxidation state = total 4s + 3d electrons. Why Fe(III) is stable: Fe³⁺ is **3d⁵, a half-filled d subshell**.

Metal	Key species and colours
V	VO₂⁺ yellow (+5) → VO²⁺ blue (+4) → V³⁺ green (+3) → V²⁺ violet (+2) — reduce with zinc in acid
Cr	[Cr(H ₂ O) ₆] ³⁺ green · CrO₄²⁻ yellow · Cr₂O₇²⁻ orange · Cr ²⁺ blue
Mn	Mn ²⁺ very pale pink · MnO ₂ brown/black · MnO ₄ ²⁻ green · MnO₄⁻ purple
Fe	[Fe(H ₂ O) ₆] ²⁺ pale green · [Fe(H ₂ O) ₆] ³⁺ yellow/brown
Co	[Co(H ₂ O) ₆] ²⁺ pink · [CoCl₄]²⁻ blue · [Co(NH ₃) ₆] ²⁺ straw/yellow
Cu	[Cu(H ₂ O) ₆] ²⁺ blue · [CuCl₄]²⁻ yellow-green · [Cu(NH₃)₄(H₂O)₂]²⁺ deep royal blue
Zn	All colourless / white

SHEET 5 — COMPLEXES, SHAPES & THE CHELATE EFFECT

Ligand: a species with a **lone pair** that forms a **dative covalent (coordinate) bond** to the central metal ion — an **electron-pair donor**; it need not be negatively charged (which is why **NH₄⁺**, with no lone pair, cannot be a ligand).

Coordination number = the number of dative bonds, NOT the number of ligands. [Ni(en)₃]²⁺ has **3 ligands but coordination number 6**; [Ni(EDTA)]²⁻ has **1 ligand, coordination number 6**.

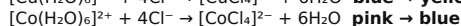
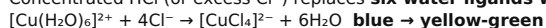
Monodentate: H₂O, NH₃, Cl⁻, CN⁻, OH⁻, CO, SCN⁻. **Bidentate:** en (NH₂CH₂CH₂NH₂), **ethanedioate** C₂O₄²⁻. **Hexadentate:** EDTA⁴⁻.

Identifying a bidentate ligand: it has **two atoms bearing lone pairs, far enough apart to bond to the same metal ion**, forming a stable five-membered ring.

Coord. no.	Shape	Bond angle	Examples
6	Octahedral	90°	Most aqua ions [M(H ₂ O) ₆] ⁿ⁺ , [Fe(CN) ₆] ³⁻ , [Ni(EDTA)] ²⁻
4	Tetrahedral	109.5°	[CuCl ₄] ²⁻ , [CoCl ₄] ²⁻ , [FeCl ₄] ⁻
4	Square planar	90°	cis-platin [Pt(NH ₃) ₂ Cl ₂], [Ni(CN) ₄] ²⁻
2	Linear	180°	[Ag(NH ₃) ₂] ⁺

THE CHLORIDE RULE — AND WHY 6 BECOMES 4

Concentrated HCl (or excess Cl⁻) replaces **six water ligands with four chloride ligands**, giving a **tetrahedral** complex:

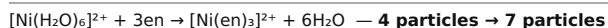


Why: chloride ions are larger than water molecules, so only four can fit around the metal ion — not enough room for six.

Isomerism: **cis-trans** in square planar MA₂B₂ (cis-platin is anticancer active, trans is not) and octahedral MA₄B₂; **fac/mer** in MA₃B₃. **Optical** isomerism when there is **no plane of symmetry** — classically an octahedral complex with **three bidentate ligands** ([Ni(en)₃]²⁺, [Cr(C₂O₄)₃]³⁻), or a tetrahedral complex with four different ligands.

Drawing: octahedral — four in the plane (two normal, one wedge, one dashed) plus one above and one below · tetrahedral — two normal, one wedge, one dashed · **always show the overall charge outside the square bracket.**

THE CHELATE EFFECT — A GUARANTEED EXPLANATION (3 MARKS)



1. An **increase in the number of particles**, so a **large positive ΔS_{system}** · **2.** the bonds broken (Ni-O) and made (Ni-N) are **similar in number and strength**, so **ΔH ≈ 0** and hence **ΔS_{surroundings} ≈ 0** · **3.** therefore **ΔS_{total} is positive**, the reaction is feasible, and the chelate complex is **more stable**.

EDTA is preferred to en because it is **hexadentate** — one EDTA replaces six waters, giving **2 → 7 particles**, an even larger entropy increase.

SHEET 6 — COLOUR (every paper, 2-6 marks — learn the three sentences exactly)

WHY COMPLEXES ARE COLOURED — THREE MARKING POINTS

1. The **ligands cause the (3)d ORBITALS to split** into two groups of slightly different energy.
2. **Visible light (a photon) is absorbed, promoting an electron from the lower to the higher energy level** (a d-d transition); $\Delta E = hf$.
3. **The light that is not absorbed is transmitted or reflected, so the complementary colour is seen.**

REJECTED: "d orbital splits" (singular) — **automatically rejected, write d orbitals** · "d-d transitions occur" **on its own**, unless you have already said the orbitals are split · "electrons **fall back down and emit** light" · "**the colour** is absorbed" — say **light/photons**.

WHY SOME ARE COLOURLESS — AND ITS OWN REJECTION

The ion has **no d electrons (d^0) or a completely full d subshell (d^{10})**, so **no electron can be promoted** between the split d orbitals — **no d-d transition**.

REJECTED: "the d orbitals **cannot be split**" (they still split — there are just no electrons to promote) · "the subshell is **empty**" for a d^{10} ion.

d^0 : Sc^{3+} , Ti^{4+} , Cr^{6+} , d^{10} : **Zn^{2+} , Cu^+ , Ag^+** .

Why $Cr_2O_7^{2-}$ is orange if Cr^{6+} is d^0 : its colour arises from **charge transfer**, not d-d transitions — the same for MnO_4^- .

WHY TWO COMPLEXES DIFFER IN COLOUR

Different ligands (or oxidation state, coordination number or shape) cause **a different degree of splitting of the d orbitals → a different energy, and hence frequency/wavelength, of light is absorbed → a different complementary colour is seen.**

REJECTED: "the number of ligands is different" or "the charge is different" **on their own** — you **must link the difference to the size of the splitting**.

Colour wheel (absorbed → seen): red → green · orange → blue · **yellow → violet/blue** · green → red/purple · **blue → orange** · violet → yellow.

Colorimetry: choose a filter of the **complementary colour to the solution** (i.e. the colour it absorbs) so **absorbance is maximised** · **zero with a blank** first · plot a **calibration curve** of absorbance against concentration with standards · read the unknown off the line. **Limitation:** readings outside the calibration range require **extrapolation**, which is unreliable.

SHEET 7 — REACTIONS OF AQUEOUS IONS & CATALYSIS

THE MOST-TESTED DISTINCTION

Adding **NaOH** or **limited NH₃** to make a precipitate is **DEPROTONATION (acid-base)**, not ligand exchange. Adding **excess NH₃** to dissolve the precipitate **is** ligand exchange.

Ion	Limited NaOH / NH ₃ → precipitate	Excess NaOH	Excess NH ₃
Cu²⁺ blue	pale blue	insoluble	deep royal blue [Cu(NH₃)₄(H₂O)₂]²⁺ — only FOUR waters replaced
Fe²⁺ pale green	green, turns brown on standing (air oxidation)	insoluble	insoluble
Fe³⁺ yellow	brown	insoluble	insoluble
Cr³⁺ green	grey-green	dissolves → [Cr(OH)₆]³⁻ green	dissolves → purple [Cr(NH₃)₆]³⁺
Zn²⁺ colourless	white	dissolves → [Zn(OH)₄]²⁻	dissolves → colourless [Zn(NH ₃) ₄] ²⁺
Co²⁺ pink	blue, then pink	insoluble	straw/yellow [Co(NH ₃) ₆] ²⁺ , browns in air

Cr(III) and Zn(II) hydroxides are AMPHOTERIC — they dissolve in **both** excess NaOH and dilute acid.

Other reagents: **SCN⁻ with Fe³⁺ → blood-red** [Fe(SCN)(H₂O)₅]²⁺ (ligand exchange) · **KI with Cu²⁺ → 2Cu²⁺ + 4I⁻ → 2CuI(s) + I₂** (white ppt in a brown solution) · **Na₂CO₃ with M³⁺ → effervescence AND a hydroxide precipitate**, because [M(H₂O)₆]³⁺ is acidic enough to react with carbonate.

Why [Fe(H₂O)₆]³⁺ is acidic: the ion has a **high charge density**, so it **polarises the O-H bonds of the coordinated water, weakening them**; a water ligand releases H⁺: [Fe(H₂O)₆]³⁺ + H₂O ⇌ [Fe(H₂O)₅(OH)]²⁺ + H₃O⁺. **M³⁺ is more acidic than M²⁺.**

CATALYSIS

Why transition metals: variable oxidation states (they can accept and donate electrons, forming intermediates) and **partially filled d orbitals** allowing adsorption.

Heterogeneous — four steps: reactants are **ADSORBED onto ACTIVE SITES → bonds weakened**, molecules held in the right orientation → reaction with a **lower activation energy** → products **DESORBED**. Fe (Haber) · V₂O₅ (Contact) · Pt/Pd/Rh (catalytic converter) · Ni (hydrogenation).

Contact process — the order matters: V₂O₅ + SO₂ → V₂O₄ + SO₃ (SO₂ reduces the vanadium first), then V₂O₄ + ½O₂ → V₂O₅.

Homogeneous: the catalyst forms an **intermediate by being oxidised (or reduced) to a different oxidation state, then reverts**. S₂O₈²⁻ + 2I⁻ catalysed by Fe²⁺ or Fe³⁺ — the same two steps in the opposite order.

Justifying by E°: the catalyst's E° must **lie between** those of the two half-cells, so **both steps have a positive E° cell**.

Autocatalysis: a product acts as the catalyst. MnO₄⁻ + C₂O₄²⁻ → Mn²⁺ **is the autocatalyst**. The rate-time graph: **slow at first** (no catalyst, two negative ions repel, high E_a) → **increases as Mn²⁺ forms**, even though concentrations are falling → **falls later** as reactants are used up. **S-shaped** product-time curve.

SHEET 8 — BENZENE & ELECTROPHILIC SUBSTITUTION

Benzene: planar regular hexagon, 120° , each carbon sp^2 hybridised with three σ bonds; the remaining **p orbitals overlap sideways** to form a **delocalised π system** above and below the plane, with **6 delocalised electrons** and six identical C-C bonds. **Mark-scheme point: the delocalised electrons come from p ORBITALS overlapping to form π bonds — not s orbitals, not σ bonds.**

THE FOUR PIECES OF EVIDENCE — A GUARANTEED 4-6 MARK QUESTION

- 1. Bond lengths (X-ray):** all six C-C bonds are the **same length, 139 pm, intermediate** between C-C (154) and C=C (134). The Kekulé structure predicts alternating long and short bonds.
- 2. Enthalpy of hydrogenation:** cyclohexene + $H_2 = -120$, so "cyclohexa-1,3,5-triene" would be $3 \times (-120) = -360$; benzene is **-208**. Benzene releases **less energy than expected**, so it is **152 kJ mol⁻¹ more stable** — the **delocalisation energy**. Always use the numbers given in the question.
- 3. Reactivity:** benzene **does not decolourise bromine water** and undergoes **substitution, not addition**, needing a catalyst. The delocalised ring has a **lower electron density than an isolated C=C**, so it **cannot polarise a bromine molecule**.
- 4. Isomer count:** only **one** 1,2-disubstituted isomer exists; Kekulé predicts two.

THE MECHANISM — AND THE FOUR ERRORS SCHEMES LIST EXPLICITLY

Step 1 generate the electrophile E^+ · **Step 2** a curly arrow **from INSIDE the ring (from the circle)** to E^+ , giving the intermediate: a **horseshoe covering AT LEAST THREE carbons (never all six)**, a **+** charge **inside or just above it**, and the **sp^3 carbon carrying BOTH H and E** · **Step 3** a curly arrow from the **C-H BOND back into the ring**, restoring delocalisation and releasing H^+ · **Step 4** regenerate the catalyst.

Rejected: **1.** an arrow starting **on or outside the hexagon** · **2.** an arrow going to a **lone pair** on the electrophile rather than to the atom · **3.** **missing the + or δ^+** on the electrophile · **4.** a horseshoe covering **all six carbons**, fewer than three, or not facing the sp^3 carbon. Also: the step-3 arrow must start from the **C-H bond, not the H atom**.

Reaction	Reagents and conditions	Electrophile
Nitration	conc. HNO_3 + conc. H_2SO_4, 50-60 °C	NO_2^+ (nitronium)
Halogenation	Br_2 (or Cl_2) with a halogen carrier — $FeBr_3$, $AlCl_3$ or Fe; anhydrous	Br^+
Friedel-Crafts alkylation	RCI + $AlCl_3$, reflux, anhydrous	R^+
Friedel-Crafts acylation	$RCOCl$ + $AlCl_3$, reflux, anhydrous	RCO^+ (acylium)
Sulfonation	Fuming H_2SO_4 , or conc. H_2SO_4 under reflux	SO_3

MCQ trap: to make **phenylethanone** from benzene you need **ethanoyl chloride with $AlCl_3$** — not ethanal, not a platinum catalyst.

Side chain: methylbenzene + Cl_2 in **UV light** substitutes in the **side chain** (free-radical); with a **halogen carrier and no UV** it substitutes in the **ring**. **Alkaline $KMnO_4$, reflux, then acidify** oxidises **any** alkyl side chain to **-COOH**.

PHENOL — AND THE WORDING THAT IS REJECTED

A lone pair of electrons ON THE OXYGEN ATOM is delocalised into the benzene ring, which increases the electron density of the ring, so it is **more easily attacked by electrophiles** and **can polarise a bromine molecule without a catalyst**.

REJECTED: "the lone pair on the O-H is delocalised" — **say the lone pair on the OXYGEN ATOM**.

Bromine water: benzene no reaction; phenol gives an **immediate white precipitate of 2,4,6-tribromophenol**, decolourising the bromine water, at room temperature with no catalyst. **Neutral $FeCl_3 \rightarrow$ purple. $NaOH(aq)$:** phenol dissolves (weakly acidic); **Na_2CO_3 : no reaction** — phenol is a weaker acid than carbonic acid.

SHEET 9 — ORGANIC NITROGEN

BASICITY — THE MOST-TESTED IDEA IN THIS TOPIC

Why amines are basic: the nitrogen has a **lone pair that can accept a proton / form a dative covalent bond with H^+** .

Order: secondary alkylamine > primary alkylamine > ammonia > **phenylamine** > amide.

Alkylamine stronger than ammonia (3 marks): **alkyl groups are electron-releasing** (positive inductive effect), so they **push electron density towards the nitrogen, increasing the electron density on N**, so the lone pair is **more available** to accept a proton.

Phenylamine weaker than ammonia (3 marks): the **lone pair on the nitrogen is delocalised into the benzene ring**, which **decreases the electron density on the nitrogen**, so the lone pair is **less available**.

Why an amide is not basic: the nitrogen lone pair is **delocalised onto the adjacent $C=O$** .

Two nitrogens (procaine): the **aliphatic** amine is the stronger base and is the one protonated by 1 mol of HCl.

Preparing amines: halogenoalkane + **excess concentrated ethanolic NH_3** , **sealed tube** (excess minimises further substitution) · **nitrile reduction** with $LiAlH_4$ in dry ether, or H_2/Ni — **adds one carbon** · **nitroarene reduction:** Sn + **concentrated HCl, reflux, then $NaOH(aq)$** (the $NaOH$ liberates the free amine from its salt) · amide + $LiAlH_4$.

Which amines cannot come from a nitrile: reduction always gives a **primary amine with $-NH_2$ on a terminal CH_2** — so not a branched, secondary or tertiary amine.

Reactions: + $HCl(aq)$ → ammonium salt ($CH_3NH_3^+Cl^-$ = methylammonium chloride) · + **HCl gas** → **white smoke** · + acyl chloride → **N-substituted amide** + **steamy HCl fumes** · + $Cu^{2+}(aq)$ → pale blue ppt, then **deep blue** in excess.

AMINO ACIDS — THE ZWITTERION EXPLAINS BOTH PROPERTIES

In the solid and at the isoelectric point: **$H_3N^+-CHR-COO^-$** . Low pH → cation; high pH → anion.

High melting temperature: amino acids exist as **zwitterions, which are ions**, so there are **strong ionic attractions between oppositely charged groups of neighbouring zwitterions**. The forces broken on melting are **ionic bonds — not hydrogen bonds, not London forces**.

High solubility: the zwitterion is **polar/ionic**, forming **ion-dipole interactions and hydrogen bonds** with water.

Glycine is NOT chiral (two hydrogens on the α -carbon); every other natural amino acid is. **Peptide link** = $-CO-NH-$; " $-COO-NH-$ " is a known MCQ trap.

Hydrolysis: reflux with 6 mol dm^{-3} HCl. **Ninhydrin** → **purple spots**.

Diazonium salts: $NaNO_2$ + **dilute HCl** (generating HNO_2 in situ) at **0–10 °C**. **Why that range — both halves needed:** **above 10 °C the diazonium ion decomposes** to phenol and nitrogen; **below about 5 °C the reaction is too slow**. **MCQ trap:** the reason is **not** that it is highly exothermic or that the ring would be nitrated.

Coupling with phenol in alkali or with an aromatic amine gives an **azo dye, para** to the OH or NH_2 .

Why azo compounds are coloured: an **extended delocalised electron system** (two rings joined through $-N=N-$) gives an **energy gap corresponding to visible light**, so part of the visible spectrum is absorbed and the **complementary colour is seen**. **Larger delocalised system → smaller gap → longer wavelength absorbed**.

Polyamides vs polyalkenes: polyamides have **polar N-H and $C=O$ groups**, so **hydrogen bonds form between the chains**; polyalkenes have only **London forces** → higher melting temperature.

SHEET 10 — SYNTHESIS, SPECTRA & THE PHRASE BANK

Grignard: RMgBr in DRY ether, reflux	Product after hydrolysis with dilute acid
CO ₂ (dry ice)	carboxylic acid RCOOH — adds one carbon
Methanal HCHO	primary alcohol
Any other aldehyde	secondary alcohol
Ketone	tertiary alcohol
Water	alkane — which is why the apparatus must be dry

Nitriles: halogenoalkane + **KCN in ethanol, reflux** → nitrile (**adds one carbon**) · carbonyl + HCN/KCN → hydroxynitrile · **acid hydrolysis:** dilute HCl, reflux → carboxylic acid · **alkaline hydrolysis** → the **carboxylate salt**, then **add acid to protonate it** — **not "to neutralise the NaOH" and not "to catalyse"**.
Key conditions: LiAlH₄ in dry ether · K₂Cr₂O₇/H₂SO₄ **distil off** (aldehyde) or **reflux** (acid) · acid → acyl chloride with **PCl₅/SOCl₂** · ester from acid + alcohol with **conc. H₂SO₄** · nitrobenzene → phenylamine with **Sn + conc. HCl, then NaOH**.

MS fragments: 15 CH₃⁺ · 29 C₂H₅⁺/CHO⁺ · 43 CH₃CO⁺ · 45 COOH⁺ · **77 C₆H₅⁺** · **91 C₆H₅CH₂⁺** · **105 C₆H₅CO⁺**. **Cl: M:M+2 = 3:1; two Cl: 9:6:1; Br: 1:1.** M+1 from ¹³C, height ≈ 1.1 × (number of C) %.

IR: O-H acid 2500–3300 **very broad** · O-H alcohol 3200–3600 broad · **N-H 3300–3500, often a double peak for a primary amine** · C≡N 2220–2260 sharp · C=O 1680–1750 · C=C 1620–1680. **Quote both the bond and the range. The absence of a peak is evidence too.**

NMR: peaks = environments (use symmetry) · areas = H ratio · **n + 1 splitting** · **O-H and N-H give a broad singlet, are not split and do not split others** (rapid proton exchange), and **disappear with D₂O**. **TMS:** inert, volatile, one sharp peak from 12 equivalent H, upfield of everything. **Solvents CDCl₃ or CCl₄ — they contain no hydrogen.**

Structure routine: M_r from MS (check Cl/Br patterns) → empirical/molecular formula → **degrees of unsaturation** (benzene ring = 4, C=O = 1, C=C = 1, ring = 1) → IR for groups **present and absent** → chemical tests → ¹³C environments → ¹H environments, ratio and neighbours → **check against EVERY piece of data.**

THE PHRASE BANK — LEARN VERBATIM

"A transition element forms **at least one stable ion with a partially filled d subshell.**"
 "The ligands cause the **d orbitals to split into two energy levels.**"
"Light is absorbed, promoting an electron from the lower to the higher energy level."
 "The light not absorbed is **transmitted/reflected — the complementary colour is seen.**"
 "A ligand **donates a lone pair of electrons to form a dative covalent bond.**"
 "There is an **increase in the number of particles, so ΔS_{system} is positive.**"
"E^ocell is directly proportional to ΔS_{total} and to ln K."
 "The reaction may still not occur because the **activation energy is very high** / the **conditions are not standard.**"
 "The **p orbitals overlap sideways to form a delocalised π system** above and below the ring."
"Benzene cannot polarise the bromine molecule, so a halogen carrier is needed."
"A lone pair on the oxygen atom is delocalised into the ring, increasing the electron density."
"Alkyl groups are electron-releasing, so the lone pair on nitrogen is more available."
 "Amino acids exist as **zwitterions**, so there are **strong ionic attractions.**"
 "Reactants are **adsorbed onto active sites, the bonds are weakened**, reaction occurs, and the products are **desorbed.**"

THE 12 MOST COMMONLY LOST MARKS

1. "d orbital" (singular) when explaining splitting. **2.** "the d orbitals cannot be split" for a d¹⁰ ion, instead of **"no electrons can be promoted"**.
3. Explaining a transition element in terms of the **atom** instead of the **ion**. **4.** Forgetting **H⁺** in an acidified half-cell. **5.** A curly arrow starting **on or outside the hexagon**. **6.** A horseshoe over **all six** carbons. **7.** Omitting the **+** or **δ+** on the electrophile. **8.** "the lone pair on the **OH**" instead of "on the **oxygen atom**". **9.** Not quoting **oxidation numbers** when proving redox. **10.** Using **HCl** instead of **H₂SO₄** to acidify a manganate(VII) titration. **11.** Confusing **deprotonation with ligand exchange**. **12.** Writing **[Cu(NH₃)₆]²⁺** instead of **[Cu(NH₃)₄(H₂O)₂]²⁺**.

Section C technique: read the passage **once, quickly**, then go to the questions — they tell you what matters. When a structure is given, **count and label the functional groups on the diagram first**. For a skeletal molecular formula, **count carbons at every line end and junction, then add hydrogens to bring each to four bonds**, and check twice. Half-life: **amount = initial × (½)ⁿ**. For "suggest" questions, **write a sensible chemical idea rather than leaving it blank**.